

RoboGeo: Benchmarking Camera-Geometry Robustness in Multi-Camera BEV Perception

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Abstract

Multi-camera bird's-eye-view (BEV) perception models are commonly trained and evaluated on a fixed camera rig. However, real deployment may involve camera-mounting changes on the same vehicle or transfer to different vehicle platforms. Existing evaluations rely on proxies rather than reproducing such camera-geometry changes. Extrinsic-only perturbation leaves the input images unchanged, whereas cross-domain generalization entangles camera geometry with domain shift. We introduce **RoboGeo**, a CARLA-based benchmark for camera-geometry robustness in multi-camera BEV perception. RoboGeo re-renders the same scenes under changed camera geometry, producing geometrically consistent image-extrinsic pairs. We define three diagnostic conditions that separate extrinsic-only perturbation, viewpoint change with base extrinsics, and viewpoint change with the corresponding extrinsics. RoboGeo provides two protocols, viewpoint perturbation and cross-platform transfer to the SUV and Bus layouts. Robustness under each is quantified by the mean Viewpoint Robustness Score (mVRS) and the Cross-Platform Transfer Score (CTS), respectively. We benchmark representative BEV segmentation and 3D object detection models, providing insights into the relationship between view transformation and camera-geometry robustness.

1. Introduction

Multi-camera bird's-eye-view (BEV) perception has become a widely used paradigm for camera-based autonomous driving, supporting BEV segmentation [1, 3, 24, 28, 29, 31, 40], 3D object detection [10, 16, 18, 20, 33, 35], and end-to-end driving [9, 11]. Because these models inform safety-critical driving decisions, they must stay reliable under the conditions met at deployment [30]. Robustness benchmarks [6, 12, 41] have so far concentrated on image-level corruptions such as adverse weather, motion blur, and sensor noise, which perturb image appearance. By contrast, robustness to changes in camera geometry remains relatively underexplored.

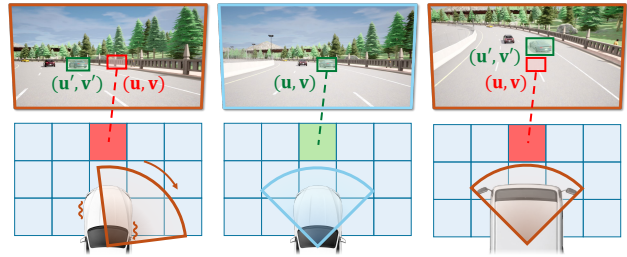


Figure 1. **Camera-geometry change breaks the BEV-image correspondence.** Under the original rig (middle), a BEV cell maps to pixel (u, v) . When the rig changes on the same vehicle (viewpoint perturbation, left) or the layout changes across vehicles (cross-platform transfer, right), the cell's content moves to (u', v') , but the model still reads from (u, v) , degrading the prediction. Green indicates the correct correspondence, and red indicates the misaligned correspondence.

Camera-geometry changes arise in two deployment settings. First, a camera-rig change occurs on a single vehicle. Localized causes such as vibration, thermal deformation, or mount loosening shift individual cameras [27, 38, 39]. Whole-vehicle motion such as acceleration, braking, or tire blow-out instead shifts all cameras together [26, 36]. In each case the perception system continues to use the unchanged extrinsics, creating an image-extrinsic mismatch. Second, a camera-layout change occurs across platforms, where a model trained on one vehicle is deployed on another whose cameras are mounted at different heights and orientations. Because re-collecting data and training for each new platform is costly, a model should ideally transfer to a new layout without retraining. In both cases, the model must handle viewpoints and camera layouts unseen during training. Since BEV models rely on camera geometry, explicitly or implicitly, to map 2D observations into BEV space, these changes misalign the resulting BEV features and degrade predictions as illustrated in Fig. 1.

Existing evaluations rely on a proxy rather than reproducing the actual camera-geometry change. Most robustness studies apply *extrinsic-only perturbation*, modify-



Figure 2. **Example images from RoboGeo.** Six-camera images re-rendered from the same scene under changed camera geometries, outlined in brown. Top: viewpoint perturbation, with a per-camera roll of the front camera and an all-camera pitch. Bottom: cross-platform transfer, with the same scene rendered from the SUV and Bus layouts.

ing the extrinsic matrices while holding the input images fixed [20, 22, 31, 35–37]. In reality, however, camera motion also alters the image itself, a *viewpoint change* that extrinsic-only perturbation does not capture, and therefore does not reflect a realistic camera-geometry change. Cross-domain generalization [14, 23, 34] changes camera layouts but also changes scene content and data distributions. The camera-geometry effect is therefore entangled with domain shift, making it difficult to attribute performance degradation to camera geometry. Recent studies have explored robustness to viewpoint and camera-configuration changes within the same domain [4, 13, 15, 19]. However, existing evaluations do not provide a unified benchmark that covers both camera-rig changes and cross-platform camera-layout changes while systematically comparing matched and mismatched image–extrinsic pairs.

We introduce RoboGeo, a benchmark for camera-geometry robustness in multi-camera BEV perception. For each camera-geometry change, RoboGeo re-renders the same scenes under the changed six-camera geometry and records the corresponding extrinsics, producing geometrically consistent image–extrinsic pairs (Fig. 2). RoboGeo provides two complementary protocols. First, the *viewpoint perturbation* protocol is designed to model camera-rig changes by applying per-camera and all-camera roll, pitch, and yaw rotations within the same vehicle platform. Second, the *cross-platform transfer* protocol deploys Sedan-trained models on the SUV and Bus camera layouts without retraining. For each geometry change, RoboGeo defines one normal condition and three diagnostic conditions that separate extrinsic-only perturbation, viewpoint change with base extrinsics, and viewpoint change with the corresponding changed extrinsics. We quantify robustness under the two protocols with the mean Viewpoint Robustness Score (mVRS) and the Cross-Platform Transfer Score (CTS), respectively. Using this diagnostic setup, we evaluate repre-

sentative BEV segmentation and 3D object detection models and report four findings. First, extrinsic-only perturbation is not a proxy for viewpoint robustness, underestimating true degradation. Second, correct extrinsics recover yaw across methods but pitch and roll only for backward projection. Third, depth-supervised forward projection collapses under the cross-platform layout shift. Fourth, recalibrating to the target extrinsics mainly benefits backward projection.

Our contributions are summarized as follows:

- We introduce RoboGeo, to our knowledge the first benchmark to cover both camera-rig and cross-platform changes for camera-geometry robustness in multi-camera BEV perception, spanning 212 camera geometries. We will publicly release the dataset.
- We define one normal and three diagnostic image–extrinsic conditions that separate a real viewpoint change from an extrinsic-only perturbation and isolate the recovery from recalibration, and propose two protocol-level metrics, mVRS and CTS.
- Evaluating six 3D detection and seven BEV segmentation models, we find that extrinsic-only perturbation underestimates the degradation from viewpoint change, and that both recalibration recovery and cross-platform transfer depend on the view-transformation method.

2. Related Work

2.1. Viewpoint Robustness in BEV Perception

Prior work has evaluated viewpoint robustness in BEV perception by perturbing the camera extrinsic matrices at inference while holding the input images fixed, an extrinsic-only perturbation [20, 22, 31, 35–37]. In particular, DSPE [31] and CAPE [35] report improved robustness under this setting. However, real tilt or mounting errors change the captured image, so evaluations that perturb only the extrinsics reproduce the effect only indirectly. Recent work [4, 13]

instead releases data with changed viewpoints. Klinghoffer et al. [13] provide a single-camera CARLA dataset of viewpoint changes for BEV segmentation, and VR-Drive [4] renders viewpoint-changed nuScenes data with 3D Gaussian Splatting for end-to-end driving. Both supply viewpoint-changed images, but each is limited, one to a single camera and the other to novel-view synthesis of real scenes. Unlike a simulator, novel-view synthesis reconstructs the scene from images and introduces rendering artifacts and reduces details, a gap that CamShift [8] measures, finding that neural-rendered views lower detection accuracy relative to directly simulated views. RoboGeo renders every viewpoint and platform directly in the simulator across all six cameras, avoiding this novel-view-synthesis gap.

2.2. Cross-Configuration Generalization

Deploying BEV perception models across vehicles requires generalization to camera placements that differ in mounting height, spacing, and orientation. Cross-domain generalization [5, 14, 23, 32, 34] evaluates this across real datasets, but transferring between datasets entangles camera geometry with scene content, sensor characteristics, and data distributions. Other work studies cross-configuration changes within the same domain. UniDrive [19] considers changes in camera configuration such as camera count and intrinsics. CamShift [8] generates sedan and SUV platforms in CARLA and uses neural rendering to adapt a detector across them. CHARM3R [15] focuses on unseen camera heights for monocular 3D detection. Each addresses only part of the cross-configuration problem (Tab. 1), and no benchmark unifies per-camera viewpoint perturbation and cross-platform transfer within a single multi-camera setup.

RoboGeo fills this gap. By re-rendering the same scenes while varying only the multi-camera geometry, RoboGeo adds the per-camera viewpoint perturbation that prior studies omit and evaluates transfer across three platforms, the Sedan, SUV, and Bus.

3. RoboGeo

3.1. Benchmark Design

With RoboGeo, we evaluate the camera-geometry robustness of multi-camera BEV perception models. Throughout the paper, we refer to the cameras mounted on a vehicle as its camera *rig*, to their platform-specific placement as its camera *layout*, and to the canonical Sedan configuration as the *base* configuration. A camera-rig change keeps the same platform and rotates one or more cameras about their mounts, so only camera orientation varies. A camera-layout change instead moves to a different platform and remounts all six cameras, changing height, spacing, and orientation together. For both BEV vehicle segmentation and 3D object detection, each model is trained only on the base configuration and

Dataset	Synth.	#Cam	#Geo	Layout	Rotation			
					R	P	Y	Intr.
Klinghoffer [13]	C+NVS	1	42	✓		✓	✓	
VR-Drive [4]	NVS	6	5	✓		✓		
UniDrive [19]	C	4–8	8	✓				✓
CamShift [8]	C	6	2	✓				
CHARM3R [15]	C	1	11	✓				
RoboGeo (Ours)	C	6	212	✓	✓	✓	✓	✓

Table 1. **RoboGeo versus existing camera-geometry datasets.** #Cam is the camera count (a range means it is varied) and #Geo the number of distinct camera geometries each dataset evaluates. The Rotation (R: roll, P: pitch, Y: yaw), Layout (camera height and platform), and Intrinsic columns mark which factors each dataset varies. Synth. is the data synthesis method, either CARLA (C) or novel-view synthesis (NVS).

evaluated on unseen camera geometries without retraining. RoboGeo provides two protocols, *viewpoint perturbation* and *cross-platform transfer* (Sec. 3.3), under the image-extrinsic conditions of Sec. 3.4.

3.2. Data

RoboGeo is built on SimBEV [25], a data generation tool based on CARLA [7]. We construct the benchmark by systematically varying camera geometry over a shared set of driving scenes. For each geometry, CARLA geometrically re-renders the six multi-view images with their corresponding extrinsics. Intrinsics are held fixed throughout, leaving the camera pose as the only quantity that varies. We convert the annotations into the nuScenes [2] schema. On top of this, we generate object attributes and a four-level visibility that measures how much of each object is visible across the cameras, computed per platform from depth and the 3D bounding boxes. Because occlusion depends on the camera mounting geometry [37], this visibility varies across platforms and quantifies how much each platform observes. The scenes inherit the diverse towns, weather, and traffic of SimBEV, giving a varied scene and object distribution.

RoboGeo comprises 268 driving scenes, split into 220 for training and 48 held out for evaluation. We render the full training and evaluation splits at 5 Hz for all three platforms (Sedan, SUV, and Bus), giving 17,380 training frames and 3,792 evaluation frames per platform. For viewpoint perturbation, we render the 210 geometries produced by the per-camera and all-camera rotations. Together with the SUV and Bus layouts, RoboGeo spans 212 distinct camera geometries beyond the base configuration, all derived from the same underlying scenes.

3.3. Geometry-Change Protocols

Viewpoint perturbation. This protocol rotates cameras of the base rig to model camera-rig changes. Starting from the base rig, we rigidly rotate either a single camera or all six cameras about their mount centers along the roll, pitch,

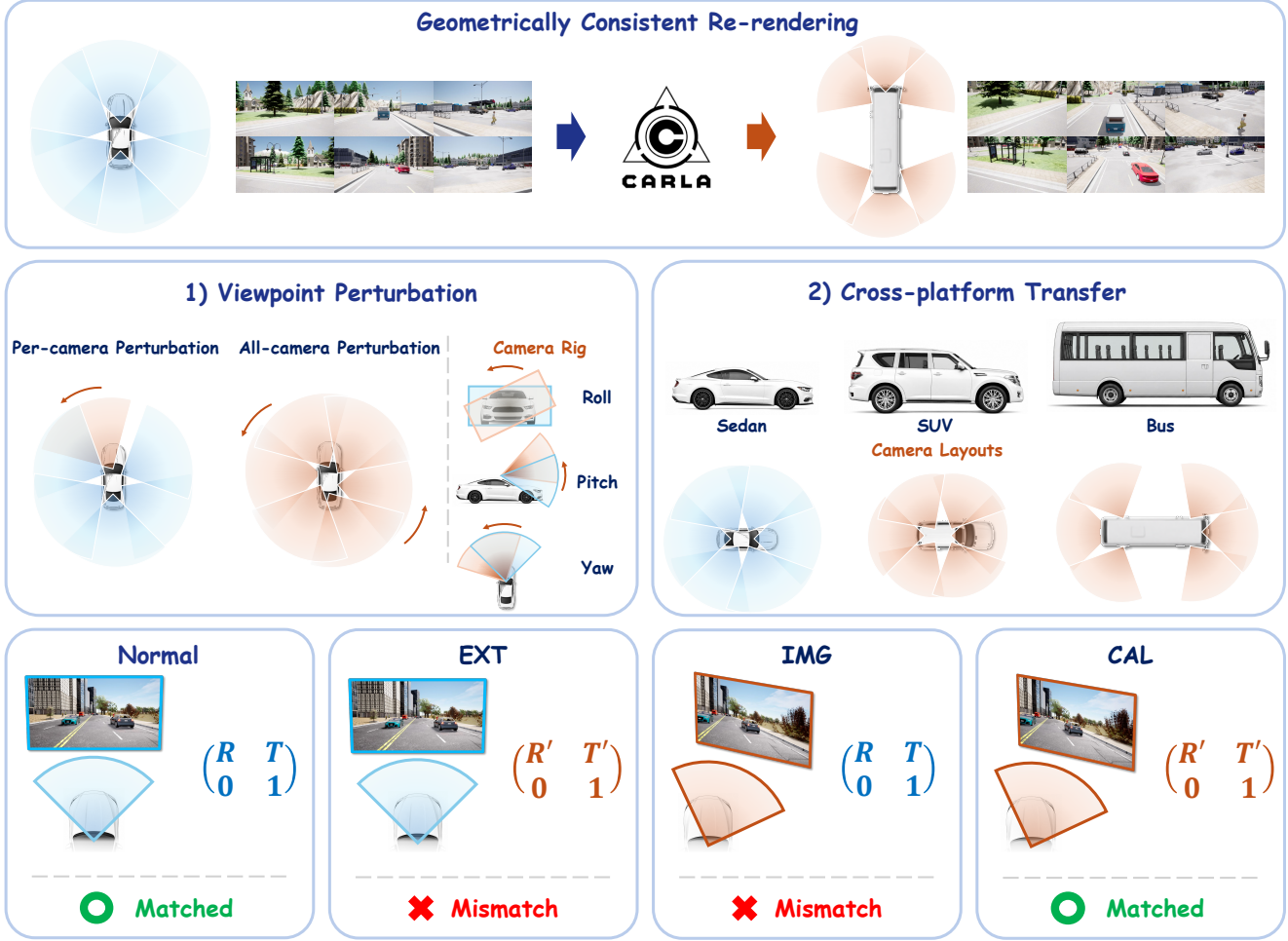


Figure 3. **RoboGeo benchmark design.** Throughout, **blue** indicates the base camera geometry and **brown** the changed geometry. (Top) Given the base rig, CARLA re-renders the same scene under each changed camera geometry, producing geometrically consistent image–extrinsic pairs. (Middle) Two geometry-change protocols. *Viewpoint perturbation* rotates a single camera or all six cameras of the base rig about the roll, pitch, or yaw, and *cross-platform transfer* deploys the Sedan-trained model on the SUV and Bus layouts. (Bottom) One normal and three diagnostic conditions. **NORMAL** and **CAL** use matched image–extrinsic pairs, whereas **EXT** and **IMG** intentionally mismatch them.

or yaw axis. We adopt the per-axis rotation range of [13] and extend it from a single camera to the full six-camera rig. Starting from the base rig, we rigidly rotate either a single camera or all six cameras about their mount centers along the roll, pitch, or yaw axis. Each axis is perturbed at $\theta \in \{\pm 4^\circ, \pm 8^\circ, \pm 12^\circ, \pm 16^\circ, \pm 20^\circ\}$, yielding 30 single-axis perturbations in total. We apply these rotations under two modes, per-camera and all-camera. The **per-camera** mode perturbs one of the six cameras, front-left (FL), front (F), front-right (FR), back-left (BL), back (B), or back-right (BR), while the other five stay unchanged. This mode models changes localized to a single camera, and we repeat the procedure for each of the six cameras. The **all-camera** mode applies the same rotation to all six cameras simultaneously, producing a coordinated viewpoint change in which no camera retains its *base*.

Cross-platform transfer. This protocol replaces the base layout with platform-specific camera layouts for the **SUV** and **Bus** (Fig. 3). The target platforms differ from the Sedan in their chassis, resulting in distinct camera layouts, and each platform is rendered using its own layout. Cross-platform transfer remounts all six cameras to the platform-specific layout of the target vehicle, changing the full layout rather than a single orientation.

3.4. Evaluation Conditions

The bottom row of Fig. 3 illustrates one normal and three diagnostic image–extrinsic conditions. The *changed* camera geometry is a per-camera or all-camera rotation of the base rig under viewpoint perturbation, and the SUV or Bus layout under cross-platform transfer.

These conditions pair images and extrinsics as follows. **NORMAL** uses base images with base extrinsics, the same

configuration the models are trained on. EXT uses base images with changed extrinsics, reproducing the extrinsic-only perturbation evaluated in prior work [20, 22, 31, 35–37]. IMG uses changed images with base extrinsics. CAL uses changed images with their corresponding extrinsics. We analyze these conditions through two gaps. The EXT–IMG gap contrasts extrinsic-only perturbation with actual viewpoint change, and the IMG–CAL gap measures the effect of supplying the correct extrinsics for the viewpoint change.

The primary condition follows from the situation each protocol models. In viewpoint perturbation, the change may go undetected, and the system keeps using the base extrinsics. IMG is therefore the deployed condition, while CAL shows how much perfect recalibration would recover. In cross-platform transfer, the change is intentional, and the target vehicle can be calibrated before deployment. CAL then measures transfer with the target extrinsics, while IMG probes the stricter calibration-free setting. EXT corresponds to no real deployment situation. We retain EXT only under viewpoint perturbation, as a comparison with prior extrinsic-only perturbation evaluations.

3.5. Performance metrics

We evaluate task performance with each task’s standard metric. For BEV segmentation, we report Intersection-over-Union (IoU) between the predicted BEV vehicle mask $\hat{M}$, thresholded at 0.5, and the ground-truth mask M :

$$\text{IoU} = \frac{|\hat{M} \cap M|}{|\hat{M} \cup M|}. \quad (1)$$

For 3D object detection, we adapt the nuScenes Detection Score (NDS) [2] to the CARLA domain as the CARLA Detection Score (CDS). Following NDS, we use five true-positive error types, translation, scale, orientation, velocity, and attribute, abbreviated ATE, ASE, AOE, AVE, and AAE. For each error type TP, we average its per-class value TP_c over the six SimBEV [25] classes $\mathbb{C}$ to obtain the mean error:

$$\text{mTP} = \frac{1}{|\mathbb{C}|} \sum_{c \in \mathbb{C}} \text{TP}_c. \quad (2)$$

We collect the five mean errors into $\mathbb{TP} = \{\text{mATE}, \text{mASE}, \text{mAOE}, \text{mAVE}, \text{mAAE}\}$ and combine them with mean average precision (mAP) into the CDS:

$$\text{CDS} = \frac{1}{10} \left[5 \text{mAP} + \sum_{\text{mTP} \in \mathbb{TP}} (1 - \min(1, \text{mTP})) \right]. \quad (3)$$

We denote the task performance by P (IoU or CDS). All reported values are averaged over the held-out validation split under the indicated condition. P_{NORMAL} is the model’s performance under the NORMAL condition.

3.6. Robustness metrics

We propose two protocol-level robustness metrics, the mean Viewpoint Robustness Score (mVRS) for viewpoint perturbation and the Cross-Platform Transfer Score (CTS) for cross-platform transfer.

Mean Viewpoint Robustness Score (mVRS). We evaluate viewpoint robustness over the perturbation grid

$$\mathcal{G} = \{\text{roll}, \text{pitch}, \text{yaw}\} \times \{\pm 4^\circ, \pm 8^\circ, \pm 12^\circ, \pm 16^\circ, \pm 20^\circ\}, \quad (4)$$

where each $(a, \theta) \in \mathcal{G}$ pairs a rotation axis a with a signed perturbation angle θ . Let $\mathcal{V} = \{\text{FL}, \text{F}, \text{FR}, \text{BL}, \text{B}, \text{BR}\}$ denote the six cameras and $\mathcal{M} = \mathcal{V} \cup \{\text{all}\}$, where $m \in \mathcal{V}$ perturbs only camera m (per-camera mode) and $m = \text{all}$ perturbs all six cameras simultaneously (all-camera mode). For an evaluation condition $c \in \{\text{EXT}, \text{IMG}, \text{CAL}\}$, let $P_c^{(m, a, \theta)}$ denote the task performance for $m \in \mathcal{M}$ and perturbation $(a, \theta) \in \mathcal{G}$. The Viewpoint Robustness Score (VRS) for each $m \in \mathcal{M}$ is

$$\text{VRS}_c^{(m)} = \frac{1}{|\mathcal{G}|} \sum_{(a, \theta) \in \mathcal{G}} \frac{P_c^{(m, a, \theta)}}{P_{\text{NORMAL}}}. \quad (5)$$

The mean Viewpoint Robustness Score (mVRS) averages over $\mathcal{M}$:

$$\text{mVRS}_c = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \text{VRS}_c^{(m)}. \quad (6)$$

Cross-Platform Transfer Score (CTS). For target platform $r \in \mathcal{R} = \{\text{SUV}, \text{Bus}\}$ and condition $c \in \{\text{IMG}, \text{CAL}\}$, let $P_c^{(r)}$ denote the performance of the Sedan-trained model evaluated on platform r without retraining. We define the Cross-Platform Transfer Score (CTS) as

$$\text{CTS}_c^{(r)} = \frac{P_c^{(r)}}{P_{\text{oracle}}^{(r)}}, \quad (7)$$

where the oracle $P_{\text{oracle}}^{(r)}$ is the performance of a model trained and evaluated directly on platform r . Each platform layout covers a different field of view and captures a different number of objects per view, so the best achievable performance differs across platforms. Normalizing by the oracle removes this platform-specific difficulty, leaving CTS to measure only the performance lost by transferring a Sedan-trained model.

4. Experimental Setup

BEV perception models. The view transformation is the module where a multi-camera BEV perception model uses camera geometry to convert multi-view image features into the BEV representation [17, 21]. Forward-projection

View Transformation	Model	Viewpoint Perturbation – mVRS (%)				Cross-platform – CTS (%)					
		Sedan				SUV			Bus		
		P_{NORMAL}	EXT	IMG	CAL	$P_{\text{oracle}}^{\text{SUV}}$	IMG	CAL	$P_{\text{oracle}}^{\text{Bus}}$	IMG	CAL
Task 1: 3D Object Detection											
Forward-projection	BEVDet [†] [10]	0.517	89.1	83.0	87.5	0.535	17.0	16.1	0.373	0.2	22.8
	BEVDepth [†] [18]	<u>0.535</u>	90.2	82.8	87.4	0.547	10.3	5.7	0.418	0.1	16.6
Backward-projection	BEVFormer [20]	0.504	83.2	83.9	93.6	0.503	37.2	<u>71.9</u>	0.547	18.0	<u>40.1</u>
	DFA3D [†] [16]	0.489	90.5	86.1	93.1	0.514	38.8	47.5	0.556	28.8	48.1
	DETR3D [33]	0.533	84.3	<u>84.1</u>	96.0	0.560	34.9	76.9	0.602	26.5	37.6
Projection-free	CAPE [‡] [35]	0.555	94.0	83.9	88.6	0.596	33.8	34.3	0.449	36.0	32.9
Task 2: BEV Vehicle Segmentation											
Forward-projection	LSS [29]	0.445	88.4	81.9	86.3	0.497	24.3	23.9	0.567	17.4	17.1
	GaussianLSS [24]	0.489	90.7	84.5	92.1	0.511	36.3	<u>40.9</u>	0.582	14.1	<u>36.6</u>
Backward-projection	BEVFormer [20]	0.455	80.2	79.6	88.2	0.473	18.2	23.0	0.542	18.8	36.8
	PointBeV [3]	<u>0.485</u>	79.7	79.4	92.6	0.510	23.2	67.5	0.544	10.3	20.5
Projection-free	CVT [40]	0.420	91.2	<u>83.8</u>	90.9	0.433	46.5	27.1	0.477	17.5	2.4
	LaRa [1]	0.454	91.1	82.5	90.0	0.490	31.0	38.5	0.535	27.7	21.5
	DSPE [‡] [31]	0.436	97.7	80.7	82.0	0.454	26.7	27.1	0.533	14.5	15.6

Table 2. **Main results on RoboGeo.** The EXT/IMG/CAL columns are normalized robustness scores in percent (mVRS for viewpoint perturbation, CTS for cross-platform transfer), and higher is better. P_{NORMAL} is each model’s unperturbed performance and $P_{\text{oracle}}^{(r)}$ the performance of a model trained and evaluated on platform r (IoU for segmentation, CDS for detection). Models are grouped by view-transformation method (the *View Transformation* column). Shaded columns mark the primary condition of each protocol, IMG for viewpoint perturbation and CAL for cross-platform transfer (Sec. 3.4), while EXT (gray) is reported for analysis only. The best and second-best per task in each primary column are shown in **bold** and underlined. [†] marks models trained with explicit depth supervision. [‡] marks models explicitly designed to be robust to extrinsic viewpoint changes, i.e., under the EXT condition.

methods scatter image features into BEV along predicted depth, backward-projection methods sample image features by projecting query-associated 3D reference points onto the image plane, and projection-free methods learn the transformation without explicit projection. We benchmark six 3D object detectors and seven BEV segmentation models that span all three view-transformation methods. For **3D object detection** we evaluate forward-projection BEVDet [10] and BEVDepth [18], backward-projection BEVFormer [20], DFA3D [16], and DETR3D [33], and projection-free CAPE [35]. For **BEV segmentation** we evaluate forward-projection LSS [29] and GaussianLSS [24], backward-projection BEVFormer [20] and PointBeV [3], and projection-free CVT [40], LaRa [1], and DSPE [31].

Training setup. All models are trained only on the Sedan split under the NORMAL condition. To isolate architectural robustness, all models share the same data and annotations, while keeping their original optimizers, schedules, and losses, with no data augmentation. Temporal models are run as single-frame variants, so camera-geometry robustness is measured per frame. Following each task’s standard convention, 3D object detection models detect all dynamic classes, using ImageNet-pretrained ResNet-50 backbones

at their standard resolutions and a ± 51.2 m range. BEV segmentation models take six images at 224×480 resolution and predict binary vehicle occupancy on a 200×200 BEV grid at 0.5 m per cell. Depth-supervised detectors additionally use CARLA-rendered depth. Each platform’s oracle is trained with the same recipe.

5. Analysis

RoboGeo probes camera-geometry robustness through two complementary protocols, viewpoint perturbation on a fixed platform (mVRS) and transfer to different camera layouts (CTS). We analyze all thirteen entries by their view-transformation method, examining viewpoint perturbation (Sec. 5.1) and cross-platform transfer (Sec. 5.2) in turn.

5.1. Viewpoint Perturbation

Under IMG, the primary condition for viewpoint perturbation, every model degrades relative to its own NORMAL. For 3D object detection, BEVDepth [18] reaches an mVRS of 82.8% and DFA3D [16] the highest at 86.1%. For BEV segmentation, PointBeV [3] reaches 79.4% and GaussianLSS [24] the highest at 84.5%. Even the most robust model in each task, DFA3D and GaussianLSS, lose more than 13% of their respective P_{NORMAL} .

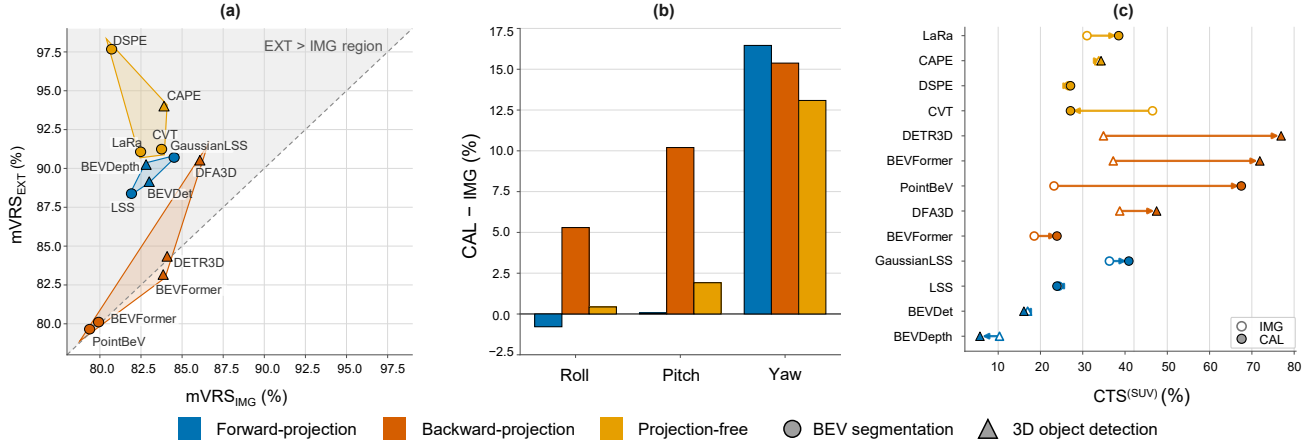


Figure 4. **Diagnostic analysis of camera-geometry robustness.** Color denotes the view-transformation method, while marker shape indicates the target task. (a) $mVRS_{EXT}$ (y -axis) vs. $mVRS_{IMG}$ (x -axis). The gray region above the diagonal marks where EXT underestimates the degradation relative to IMG. (b) Recalibration recovery (CAL - IMG) broken down per rotation axis (roll, pitch, yaw). (c) SUV cross-platform transfer, where each arrow connects IMG (open marker) to CAL (filled marker) for the same model.

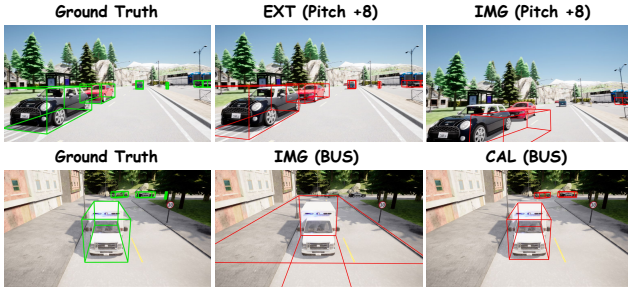


Figure 5. **Qualitative results under our diagnostic conditions.** Ground-truth boxes are green and predictions red. (Top) CAPE [35] under a +8° pitch, where EXT stays close to the ground truth while IMG degrades, showing that extrinsic-only perturbation underestimates the degradation from real viewpoint change. (Bottom) DFA3D [16] under cross-platform transfer to the Bus layout, where the degraded IMG prediction recovers under CAL. One camera is shown per panel, while predictions always use six-camera input.

Finding 1. Extrinsic robustness is not a proxy for view-point robustness. EXT and IMG measure different forms of robustness. EXT changes the extrinsics while keeping the input images fixed, whereas IMG changes the input images while keeping the base extrinsics. As shown in Fig. 4(a), EXT reports higher mVRS than IMG for most models, indicating that EXT underestimates the degradation caused by viewpoint change. The size of this gap follows the view-transformation method. Projection-free models lie farthest above the diagonal, forward-projection models are in between, and backward-projection models fall closest to it, where EXT and IMG nearly agree. Projection-free models are the most robust under EXT but lose this lead under IMG, so this extrinsic robustness does not extend to actual viewpoint change. Fig. 5(Top) illustrates this for CAPE under a pitch change, where the EXT prediction stays accurate while IMG degrades.

Finding 2. Correct extrinsics separate yaw from pitch and roll. The IMG $\rightarrow$ CAL recovery depends on both the rotation axis and the view-transformation method (Fig. 4(b)). Yaw perturbations are recovered across all three groups, indicating that yaw degradation is largely an image-extrinsic alignment error that corrected extrinsics can rectify. Pitch and roll, however, are more challenging as they alter the visible ground geometry, object scale, and vertical structure. Only backward-projection models recover them, re-sampling the image under the corrected geometry, whereas forward-projection and projection-free methods show little to no recovery.

5.2. Cross-Platform Transfer

The oracle, trained directly on each platform, varies across platforms. The Bus oracle exceeds the SUV oracle for segmentation but falls below it for forward-projection detection. Under CAL, the primary condition for cross-platform transfer, backward-projection models transfer most robustly on both platforms. On the SUV, the milder of the two shifts, the best models recover much of the oracle, reaching 76.9% for detection and 67.5% for segmentation. The Bus differs from the Sedan even more in camera height, spacing, and orientation, where even the best models reach only 48.1% for detection and 36.8% for segmentation, with most entries below half the oracle’s performance. Since the target extrinsics are already supplied, the remaining gap cannot be explained by calibration mismatch. Instead, the view transformation fails to generalize across layout shifts, and the degradation grows with the size of the shift.

Finding 3. Cross-platform transfer exposes brittle depth-supervised forward projection. Among forward-projection models, the collapse is specific to those trained

with explicit depth supervision. BEVDet and BEVDepth, both depth-supervised, are the ones that break down, whereas the forward-projection LSS [29] and GaussianLSS, trained without depth supervision, degrade but do not collapse. On SUV, BEVDet and BEVDepth recover little even with the target extrinsics, reaching only 16.1% and 5.7% under $CTS_{CAL}^{(SUV)}$. On Bus, under $CTS_{IMG}^{(BUS)}$, both collapse almost entirely, as inaccurate depth under the changed layout corrupts the resulting BEV representation. The gap is not explained by depth supervision alone, since the depth-supervised backward-projection DFA3D transfers far better, reaching 47.5% under $CTS_{CAL}^{(SUV)}$. The failure therefore arises from the combination of depth supervision and forward projection, where depth errors directly corrupt the BEV representation, rather than from either factor alone.

Finding 4. Recalibrated target extrinsics mainly benefit backward projection. On SUV, the $IMG \rightarrow CAL$ comparison shows that the benefit of target extrinsics depends strongly on the view-transformation method (Fig. 4(c)). Backward-projection models benefit the most, with every model improving. Their queries use the extrinsics to project 3D reference points onto the 2D image, so supplying the target extrinsics directly corrects the sampling. Forward-projection and projection-free models benefit far less, with several failing to improve or even degrading under CAL. On Bus, backward-projection models again improve across the board, with DFA3D recovering once the target extrinsics are supplied (Fig. 5(Bottom)). The forward-projection BEVDet and BEVDepth recover here as well, since recalibration corrects their severe IMG misalignment, but they remain among the weakest in transfer. The consistent gains from target extrinsics thus stay with backward projection.

5.3. Existing Robustness Methods

We further examine whether existing robustness methods close the gap exposed by RoboGeo. On top of the base models, we add cross-domain generalization methods (PD-BEV [23] for detection and EAFormer [34] for segmentation) and a generic extrinsic augmentation that perturbs only the training extrinsics. PD-BEV renders new viewpoints from BEV features during training to learn viewpoint-invariant features. EAFormer replaces CVT’s [40] learned positional encoding with an epipolar attention field that links each BEV cell to its epipolar line in the camera views. Because extrinsic augmentation changes the calibration but never the images, it adds little viewpoint robustness and even lowers base performance, with P_{NORMAL} dropping from 0.420 to 0.388 for CVT and from 0.535 to 0.518 for BEVDepth. The two cross-domain methods help more than augmentation, with PD-BEV raising the depth-supervised BEVDepth’s Bus transfer from 16.6% to 41.8% CTS_{CAL}

Method	P_{NORMAL}	mVRS (%)			CTS_{CAL} (%)	
		EXT	IMG	CAL	SUV	Bus
<i>BEV Segmentation</i>						
CVT [40]	0.420	91.2	83.8	90.9	27.1	2.4
+ Extrinsic Aug.	0.388	100.0	87.1	87.2	42.5	8.9
+ EAFormer [34]	0.447	92.5	85.8	91.7	49.2	11.6
<i>3D Object Detection</i>						
BEVDepth [18]	0.535	90.2	82.8	87.4	5.7	16.6
+ Extrinsic Aug.	0.518	91.0	83.9	88.1	10.8	20.3
+ PD-BEV [23]	0.561	91.4	88.9	95.1	55.9	41.8

Table 3. **Existing methods do not close the gap on either protocol.** EAFormer [34] and PD-BEV [23] are cross-domain generalization methods built on CVT [40] and BEVDepth [18], and extrinsic augmentation perturbs the training extrinsics.

and EAFormer raising CVT’s SUV transfer from 27.1% to 49.2%. However, none of these methods closes the gap on either protocol. Viewpoint robustness improves but stays short of full invariance, and cross-platform transfer remains far below the oracle. RoboGeo therefore remains far from saturated.

6. Conclusion

We introduced RoboGeo, a benchmark for camera-geometry robustness in multi-camera BEV perception. RoboGeo re-renders the same driving scenes under changed camera geometry, giving geometrically consistent image-extrinsic pairs for evaluation. On these, it defines two complementary protocols, viewpoint perturbation and cross-platform transfer, scored by the mVRS and CTS metrics. Across thirteen representative detection and segmentation models, extrinsic-only perturbation underestimates the true degradation from a real viewpoint change. Robustness also varies substantially with the view-transformation method. Existing robustness methods narrow but do not close the gap, which remains a substantial open challenge. We believe RoboGeo can establish camera-geometry robustness as a standard axis for evaluating multi-camera BEV perception and drive the design of future models that remain reliable as camera rigs and platforms change.

Limitations and future work. Constructing this study from real data is infeasible, since each driving scene occurs only once and cannot be re-recorded under different camera geometries, so evaluating all 212 distinct geometries on a shared set of scenes is only possible in simulation. This inevitably leaves an appearance gap to real-world sensors. RoboGeo also holds intrinsics, camera count, and scene content fixed and currently spans three vehicle types, leaving the joint effect of these factors, broader platform sets, and validation on real sensors to future work.

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